

Introduction to Statistical Methods

(S1-26_AIMLCZC418) – Assignment 1

Section: Prof. Amit Sharma

Each question carries 2.5 marks ($2.5 \times 4 = 10$ marks)

1. Submissions are individual.
 2. **Duration: 06th September 2026 – 30th September 2026**
 3. Solve all questions on paper, scan your handwritten solutions, and upload them.
 4. Plagiarism will result in zero marks.
 5. Write your **Name, BITS ID, and Section** on each page.
 6. Only handwritten solutions with formula, full steps, and proper justifications are required.
-

Q1.

The travel time, in minutes, taken by a student to reach college was recorded for 10 days:

42, 45, 47, 44, 46, 48, 43, 45, 47, 85

- a) Calculate the mean and median travel time.
- b) Using the Interquartile Range (IQR) method with the $1.5 \times \text{IQR}$ rule, determine whether the data contains an outlier. Show the calculation of Q1, Q3, IQR, and the outlier limits.
- c) Based on your results, state whether the mean or median gives a better representation of the typical travel time. Briefly justify your answer.

Q2.

A box contains 5 red balls, 4 blue balls, and 3 green balls. One ball is selected at random.

- a) Find the probability that the selected ball is red or blue.
- b) Find the probability that it is not green.
- c) Are the events “selecting a red ball” and “selecting a blue ball” mutually exclusive? Give a reason.

Q3.

A manufacturing company produces 70% of its components using Machine A and the remaining 30% using Machine B.

- For components produced by Machine A, the probability of a defective component is 0.02.
- For components produced by Machine B, the probability of a defective component is 0.05.

A component is selected at random from the total production.

- a) Calculate the overall probability that the selected component is defective.
- b) Given that the selected component is defective, use Bayes' theorem to calculate the probability that it was produced by Machine B.

Q4.

Let (X) represent the number of items purchased by a randomly selected customer during a visit to a store. Assume that (X) can take only the values 0, 1, 2, or 3, with the following probability distribution:

X	0	1	2	3
P(X)	0.40	0.30	0.20	0.10

- a) Verify that the given values form a valid probability distribution. State the conditions that must be satisfied and verify them using the given probabilities.
- b) Calculate the expected number of items purchased, $E(X)$.
- c) Calculate the probability that the customer purchases two or more items.